



Jaskirat Singh

Roll No.: BTECH/10177/22

Electrical And Electronics Engineering

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GitHub Profile

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EDUCATION

•BIRLA INSTITUTE OF TECHNOLOGY , MESRA

2026

B.Tech in Electrical and Electronics Engineering

CGPA: 7.15

•MUNAM PUBLIC SCHOOL,MAHARAJGANJ

2021

Central Board of Secondary Education, XII

Percentage: 95

•Angels High School, Hazaribagh

2019

Central Board of Secondary Education, X

Percentage: 90.8

TECHNICAL SKILLS AND INTERESTS

Technologies: C++, C, JavaScript, SQL

Tools/Platforms : MATLAB/Simulink, Eagle (PCB Designing), Arduino, Excel.

Interpersonal Skills: Event Management, Public speaking, Leadership, Communication, Team collaboration, Presentation skills

Areas of Interest: GenerativeAI, Power Electronics, IoT

EXPERIENCE

ProductimateAiSolutions

May 2025 - Jul 2025

Software Developer Intern

Tools & technologies: **React.js, TypeScript, Tailwind CSS, REST APIs**

- Developed a **B2B SaaS Sales Agent** platform using React.js and TypeScript, designing reusable, scalable UI components for enterprise workflows.
- Implemented **role-based authentication** and secure login/signup flows with **persistent session management**, improving platform security and user access control.
- Integrated **RESTful backend APIs with the frontend**, ensuring secure data exchange and optimized response handling across core application features.

PERSONAL PROJECTS

•Universal Battery Charger For EVs :

Dec 2024

Tools & technologies used: **Eagle CAD, Forward Converter-Based SMPS Design, High-Voltage Testing**

- Designed and developed a **forward converter-based** battery charger PCB using Eagle software
- Conducted high-voltage testing up to **300V, including voltage measurement across MOSFET drain and source.**
- Designed and implemented a protection circuit for high-voltage applications to ensure system safety and reliability.
- Complete testing and validation process, using oscilloscopes, multimeter and other instruments to analyze circuit performance, verify specifications, and ensure operational reliability.

•Power Factor Correction Device:

May 2025

Tools & technologies used: **MATLAB/Simulink, Boost Converter-Based PFC Design**

- Simulated and analyzed a **power factor correction (PFC)** system in MATLAB/Simulink using a **boost converter topology** with ICE3PCS01G IC as the digital controller model.
- Achieved improved power factor and **reduced total harmonic distortion (THD)** through simulation results, validating the design's effectiveness in enhancing overall energy efficiency.

ACHIEVEMENT

- Secured **3rd** position among 30+ teams in the “**CodeZilla**” coding competition, held as part of Pantheon 2024.
- **Core team member** for Bitotsav, leading event planning and execution for one of BIT's biggest cultural fests.

POSITION OF RESPONSIBILITY

•Project and Tech Coordinator, EEESOC

-Managed and organized hands-on workshops on domains such as **Micro-controller programming, PWM** etc.

• Tech And Management Lead - Team Aveon Racing

- Managed a team in developing the complete electrical powertrain for an electric all terrain vehicle competing in EBAJA SAEINDIA 2025.

CODING PROFILE

• **Leetcode :** | 300+ Problems Solved| Handle : **Jaskirat-singh**

• **GeeksforGeeks :** | Handle : **jaskiratsi2k1r**